

IBM BOB AI INNOVATION HACKATHON 2026 · AI TRACK

Drug Safety Signal Detector

& Regulatory Submission Readiness Checker

Two regulatory bottlenecks — signal screening and dossier readiness — in one explainable, offline tool.

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Local Streamlit prototype — not a certified PV or CTD compliance system

THE PROBLEM

Too much safety data, too little review capacity

2M+

adverse-event & medication-error reports reach FDA every year

30M+

cumulative records in FAERS — far past manual screening

5

ICH M4 CTD modules; one missing section can stall a filing



Manual signal hunting is unreliable

Spreadsheet triage across thousands of drug–event pairs misses rare combinations and is never reproducible twice.



Dossier gaps surface too late

Teams discover missing CTD sections during submission review, when rework is most expensive.



Data can't leave the building

Patient-level case data blocks most cloud AI tools, so small teams get no tooling at all.

The data got faster. The review didn't.



FAERS now publishes daily

FDA moved adverse-event publication from quarterly to daily as part of its data modernization push. Reviewers face a continuous stream instead of four batches a year.



Screening must be continuous

A quarterly spreadsheet ritual cannot consume a daily feed. Disproportionality screening has to become a repeatable, scripted step.



Explainability is non-negotiable

Regulatory reviewers must be able to reproduce a number by hand. A black-box score is not defensible in a safety review.

One app, two reviewer workflows



01 · Signal Detection

- Upload and clean a case-level CSV
- Build drug–event pair counts
- Compute a 2×2 PRR for every pair
- Rank and flag potential signals
- Optional clustering for pattern review



02 · Submission Readiness

- Map a dossier outline to Modules 1–5
- Score completeness per module
- Weight into one overall readiness score
- Classify gaps High / Medium / Low
- Export a reviewer-ready gap report

Rule-based explanations throughout. No external AI API call at runtime — the app runs fully on the reviewer's machine.

PRR screening you can recompute by hand

$$\text{PRR} = [a / (a + b)] \div [c / (c + d)]$$

Proportional Reporting Ratio — the standard first-pass disproportionality measure in pharmacovigilance.

	Event present	Event absent
Target drug	a	b
All other drugs	c	d

a = target drug with the event · b = target drug, other events · c and d = the same split across every other drug in the file.

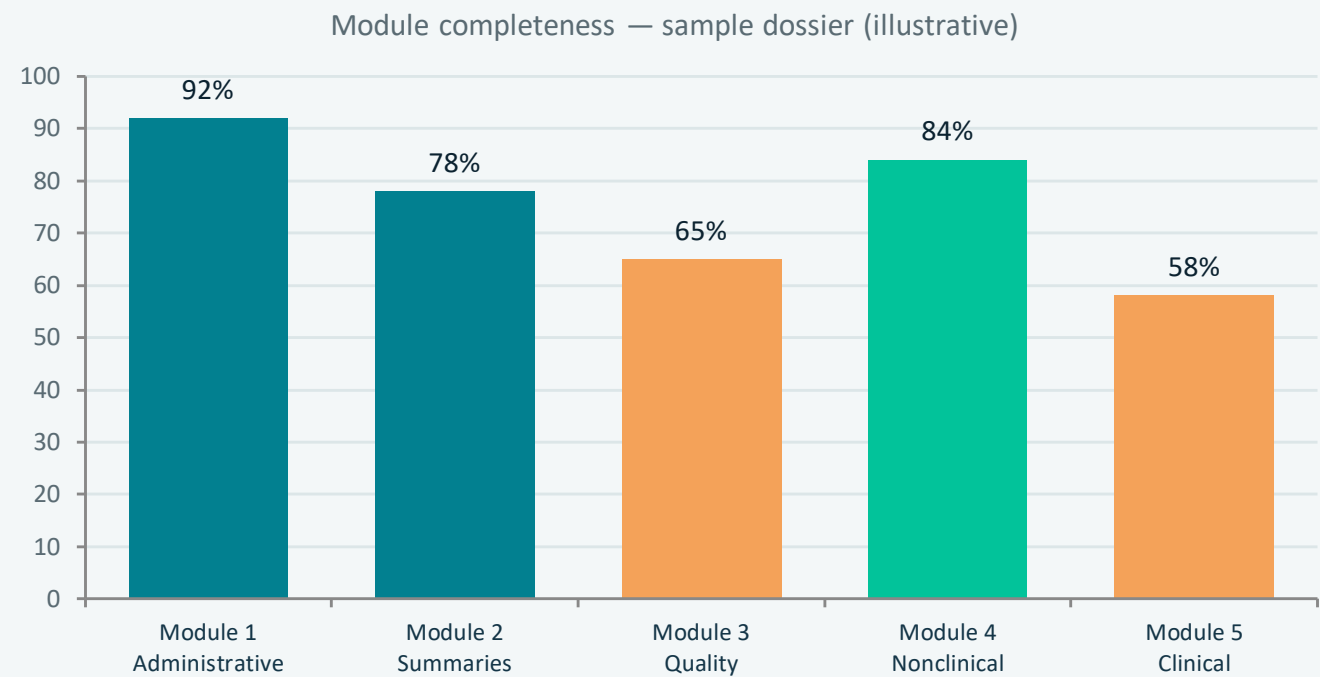
Engineering care

- Zero and empty margins handled explicitly — no divide-by-zero crashes
- Minimum case-count threshold before a pair is eligible to be flagged
- Deterministic output: same CSV in, identical ranking out, every run
- Each flag ships with its a/b/c/d counts, so a reviewer can verify it manually

Output label: “Potential Safety Signal”

A statistical flag for human follow-up — never a medical or causal conclusion.

Module-by-module scoring with a ranked gap list



Gap report — exported as CSV

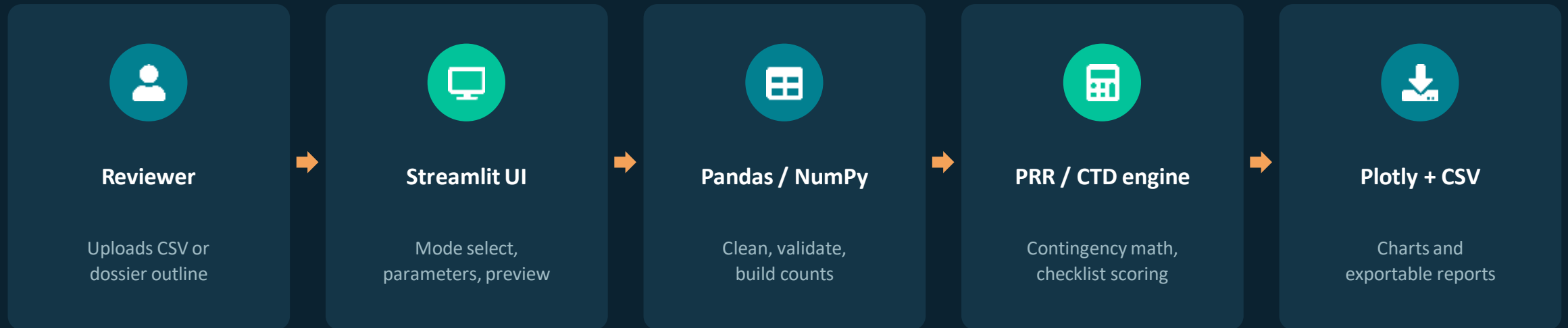
Module	Status	Priority
3 · Quality	Missing	High
5 · Clinical	Partial	High
2 · Summaries	Partial	Medium
1 · Admin	Complete	Low

Weighted overall readiness score: 74%.

Every row carries a plain-language recommendation, so the gap list reads as a work plan rather than a warning.

Checklist is a representative subset of ICH M4 CTD structure — values shown are illustrative sample output, not a real dossier.

A short, auditable pipeline



IBM Bob was used for the development workflow only — planning, code generation, tests and documentation. It is not called at runtime, and no case data ever leaves the machine.

What you'll see in 90 seconds

1

Launch locally

`python -m streamlit run src/app.py` opens at localhost:8501 — no keys, no network.

2

Load a case CSV

Column mapping and validation run instantly; malformed rows are reported, not silently dropped.

3

Screen for signals

Ranked PRR table with counts, plus a Plotly view of the top drug–event pairs.

4

Switch to readiness

Paste a dossier outline; modules score live against the CTD checklist.

5

Export

Download the ranked signal table and the prioritised gap report as CSV.

WHY THIS WINS

Four choices that separate us



Explainable by construction

Every flag shows the counts behind it. A reviewer can reproduce our number on paper — no trust required.



Private and offline

Zero runtime API calls means sensitive case data stays local. That is what makes it deployable in a real PV team.



Reproducible, not vibes

Deterministic pipeline: same input, same ranking. Screening becomes an auditable step, not an analyst's judgement call.



Two workflows, one tool

Safety screening and submission readiness land on the same team. Joining them removes a handoff instead of adding another dashboard.

Built for the edge cases, not the happy path



Zero-count cells

Pairs with $a = 0$ or $c = 0$ return a defined state instead of NaN or a crash.



Empty margins

Rows with no comparator exposure are excluded from ranking and reported separately.



Malformed CSVs

Missing columns, mixed date formats and duplicate case IDs are surfaced to the user.



Rare-pair noise

A minimum case count suppresses spurious high PRRs from one- or two-case pairs.



Repeat runs

Identical input produces byte-identical exports — verified across runs.



Checklist coverage

Every CTD checklist item maps to exactly one module and one priority band.

What this tool is — and is not



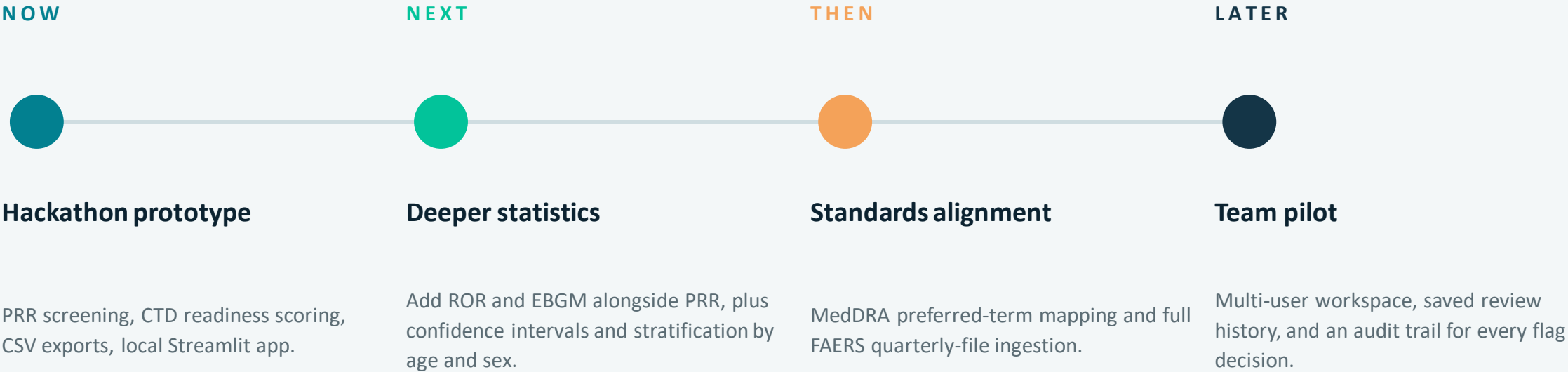
- A first-pass screening and triage aid
- A reproducible PRR calculator with visible inputs
- A representative CTD readiness checklist
- A prototype that runs entirely offline



- A replacement for pharmacovigilance experts
- A certified PV or CTD compliance system
- A causal or medical conclusion about any drug
- Complete FAERS coverage — sample data is synthetic

Stating these limits is part of the design. In a regulated domain, a tool that overclaims is a tool that cannot be adopted.

From prototype to pilot



TEAM TECHTRESS

Faster screening. Fewer surprises at filing.

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Lead

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Member

```
python -m streamlit run src/app.py
```

→ <http://localhost:8501> · runs offline, no API keys